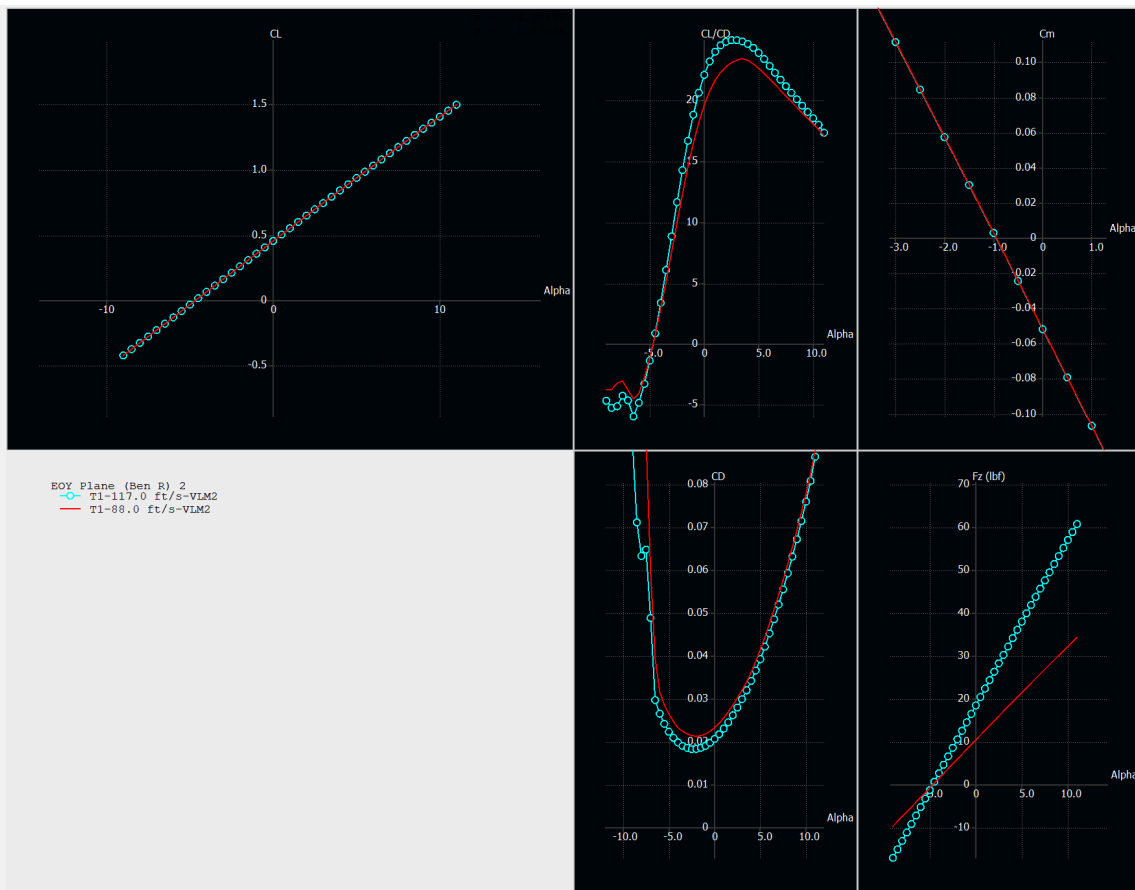
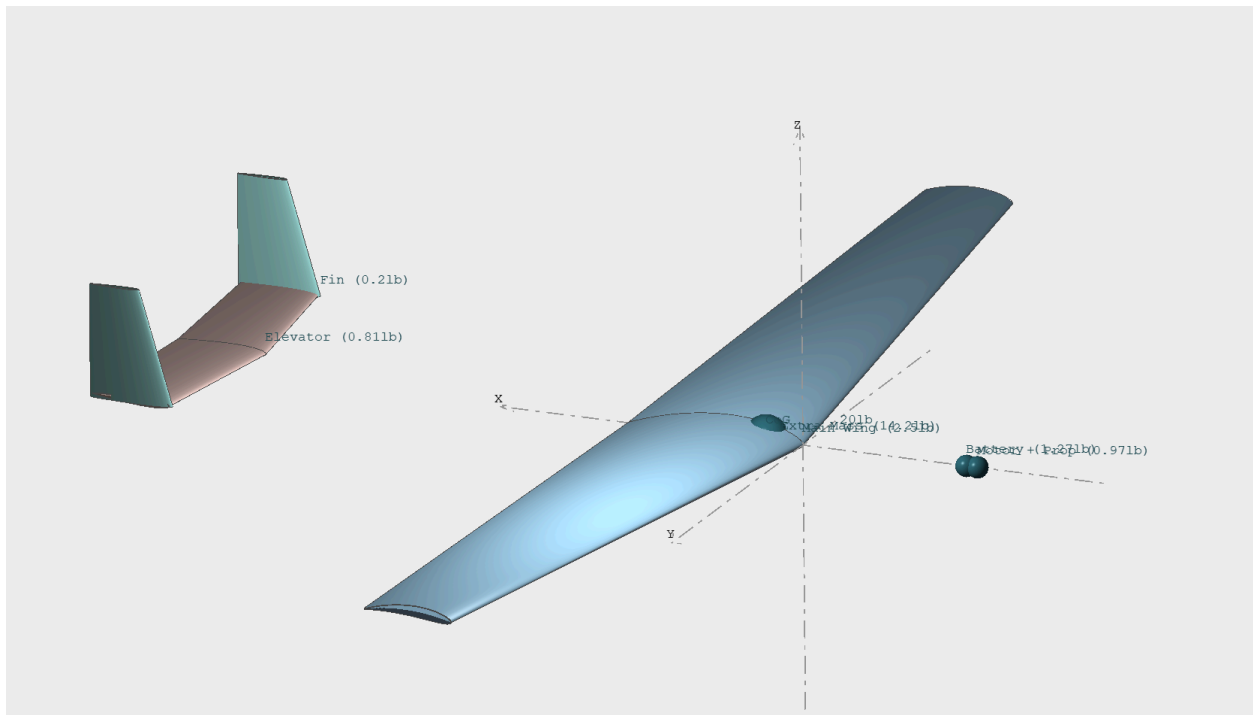
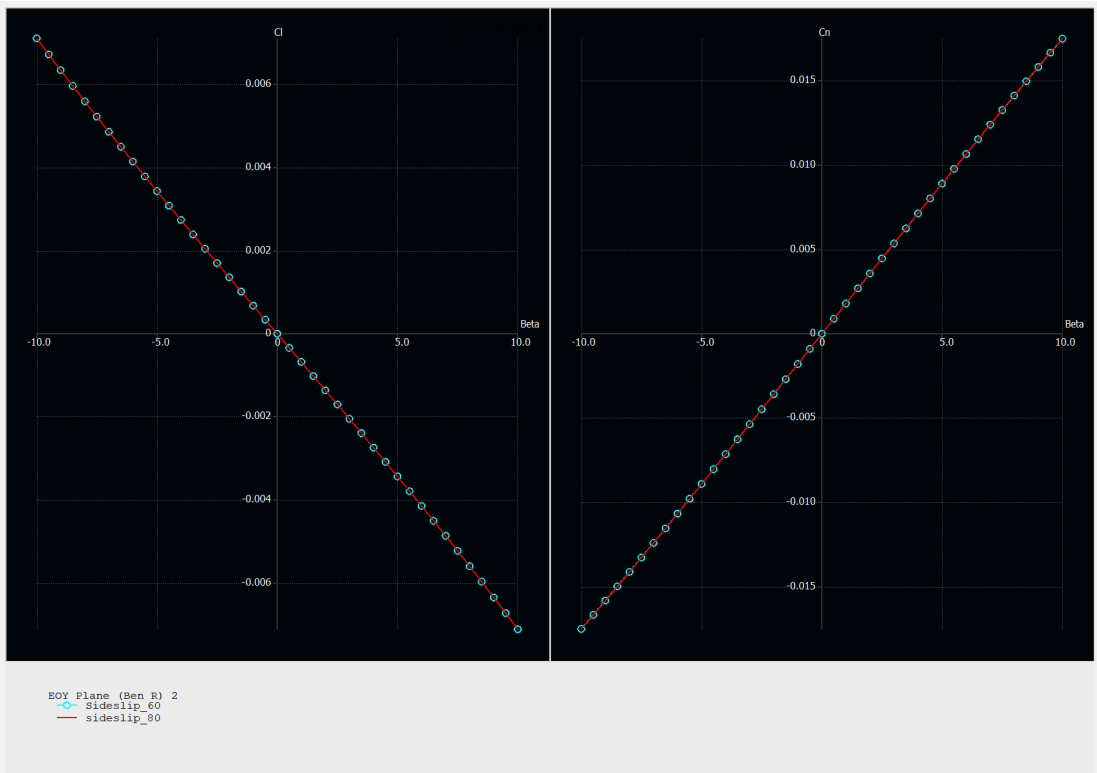
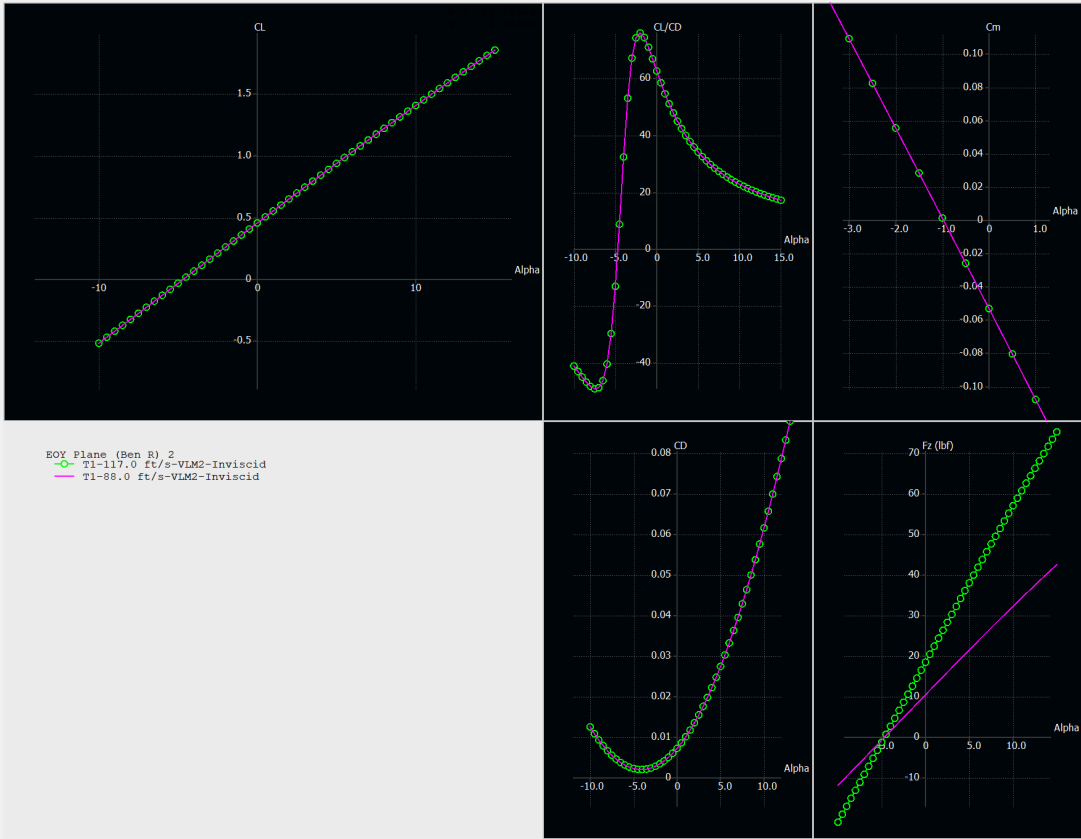
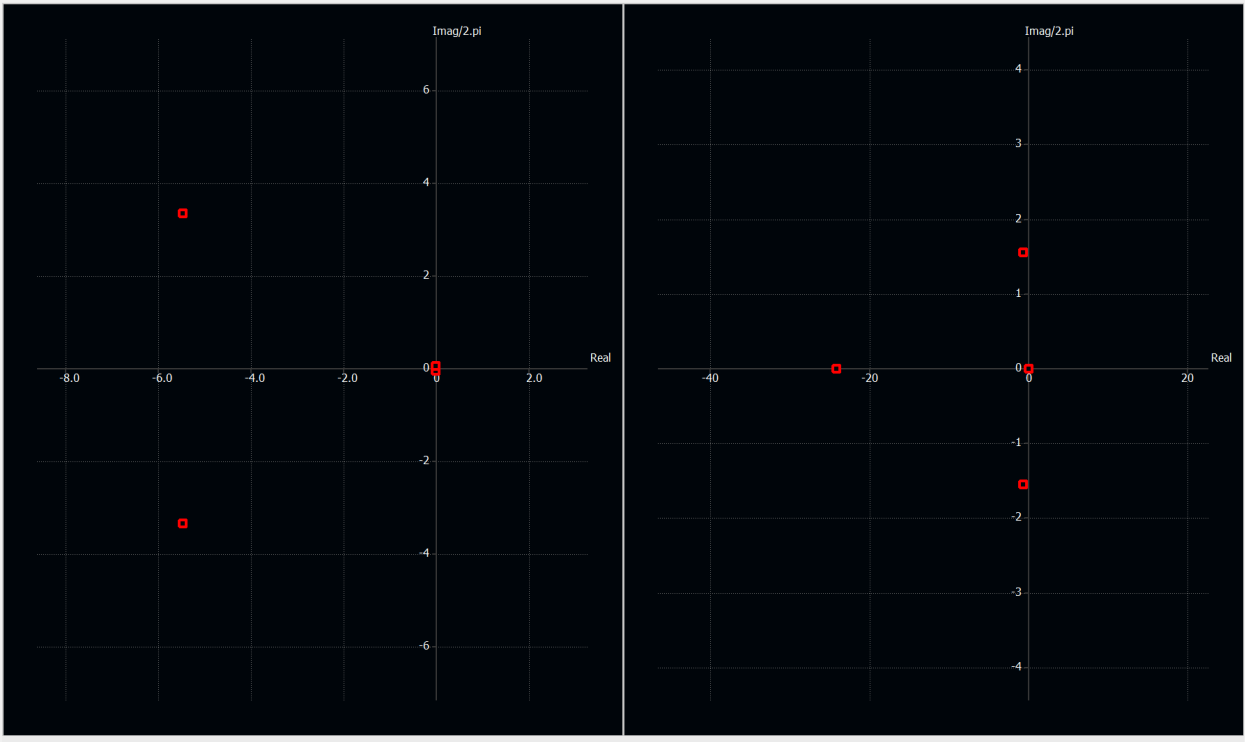


Ben Roth XFLR5 Assignment

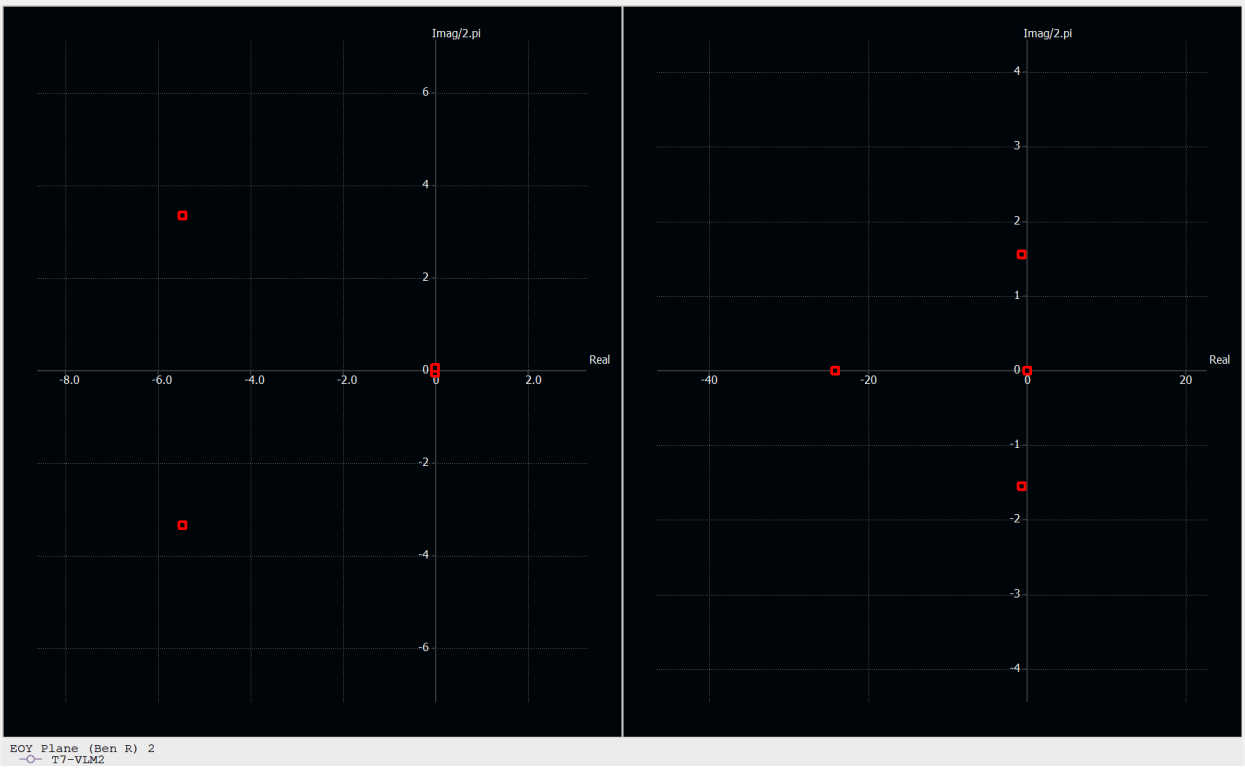




Longitudinal:



Lateral:



Design Choices and Analysis:

- Most years, the DBF competitions tend to be rather similar every year, maximizing efficiency in three categories: speed, carrying capacity, and maneuverability. My design emphasizes these. The biggest thing I implemented with this design was an H-tail (twin fin) for improved stability. These vertical stabilizers on the ends of the horizontal stabilizers act as a barrier, blocking high-pressure air underneath the stabilizer, reducing tail vortices. In addition, this design allows for smooth, clean air to hit the V-stab, allowing for more rudder authority, control, and yaw dampening. As shown by the C_l vs. β and C_n vs. β graphs, the plane is directionally stable. Furthermore, I made the tail section with a tail volume of 0.63 to allow for increased control and maneuverability, especially when carrying large amounts of mass. For the wing, I made it a higher aspect ratio wing, with a slight sweep of 4.29 degrees, and 2 degrees of dihedral. I did this to reduce induced drag and allow for more efficient lift production, allowing the plane to fly longer on less battery, like a glider. For the wing, I took inspiration from the B29 and B24, where they have high aspect ratio wings, and this allows them to travel far, while carrying large amounts of weight efficiently, while also allowing the aircraft to have high roll stability. I also made the taper ratio 0.5, as this also reduces induced drag and allows the rolling moment to be reduced, where less work has to be done to roll the aircraft. Finally, the CG was placed approximately at 20% MAC, which is not too far forward, allowing for maneuverability, because if it were too far forward or too far back, the plane would become too stable or unstable (respectively). I also added a -1.5 degree twist to the wing and set the elevator to -2 degrees to improve the stability of the aircraft.
- Reviewing all the data collected, the aircraft is in a stable condition. Under viscous conditions, the aircraft was stable until about 11.5 degrees, where the wing encountered an aerodynamic stall, causing the program to stop for the NACA6512 airfoil. To get the full range of AOA, I ran an analysis in Inviscid, and this gave me the full range (-10 to 15) as depicted in the graphs above. Since this is a heavy cargo plane, it makes sense that the stall AOA is around 11 to 12 degrees, but this can be improved by adding vortex generators, increasing the chord of the wing by a few inches, or even adding slats or flaps. At a cruise speed of 60 and 80 mph, the plane flew great with only a 1-2 degrees of needed trim for perfect flight. For directional stability, the plane is capable of restoring itself, given weather conditions or disruptions in cruise. The dihedral also helps with keeping the aircraft level during a slip. Dynamically, the Type 7 root locus analysis confirms stability, as all primary oscillatory modes consist of complex eigenvalue pairs located securely within the negative left-hand plane. In the longitudinal axis, the high-frequency Short Period mode is heavily damped by the large horizontal stabilizer, instantly suppressing sudden pitching transients, while the lightly damped Phugoid mode manifests as a slow, easily manageable altitude-to-speed exchange. Laterally, the airframe shows stability through its Roll Damping mode, where a deeply negative real root ensures rapid roll cessation the moment control sticks are neutral. In addition, the Dutch Roll complex pairs land safely at Real = -0.69, proving that the twin vertical

stabilizers of the H-tail provide sufficient directional damping to rapidly arrest any side-to-side tail-wagging oscillations and ensure a predictable, stable flight profile.